



ST BENEDICT'S

SUBJECT	Science: Paper 2	DATE	19 July 2018
GRADE	11	MARKS	130
EXAMINER	Mrs Angelova	MODERATOR	Mr McCoy
NAME		DURATION	120 minutes
CLASS			

**LOW
ORDER**

55%

HIGH ORDER

45%

INSTRUCTIONS

1. This examination consists of 11 pages and a separate answer sheet.
2. A data sheet has been provided.
3. It is in your own interest to write legibly and to present your work neatly.
4. Answer all questions on the folio paper unless instructed to do otherwise.
5. Start each question on a new page.
6. Record all answers to three significant figures.
7. Rough work and planning may be done as long as it is clearly indicated as rough work. Rough work will not be marked.

Question	1	2	3	4	5	6	Total
Marks	12	23	25	27	17	26	130

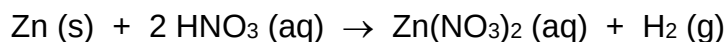
QUESTION 1: MULTIPLE CHOICE

12 MARKS

On the **answer sheet** make a cross (X) in the box corresponding to the letter which you consider to be the most correct.

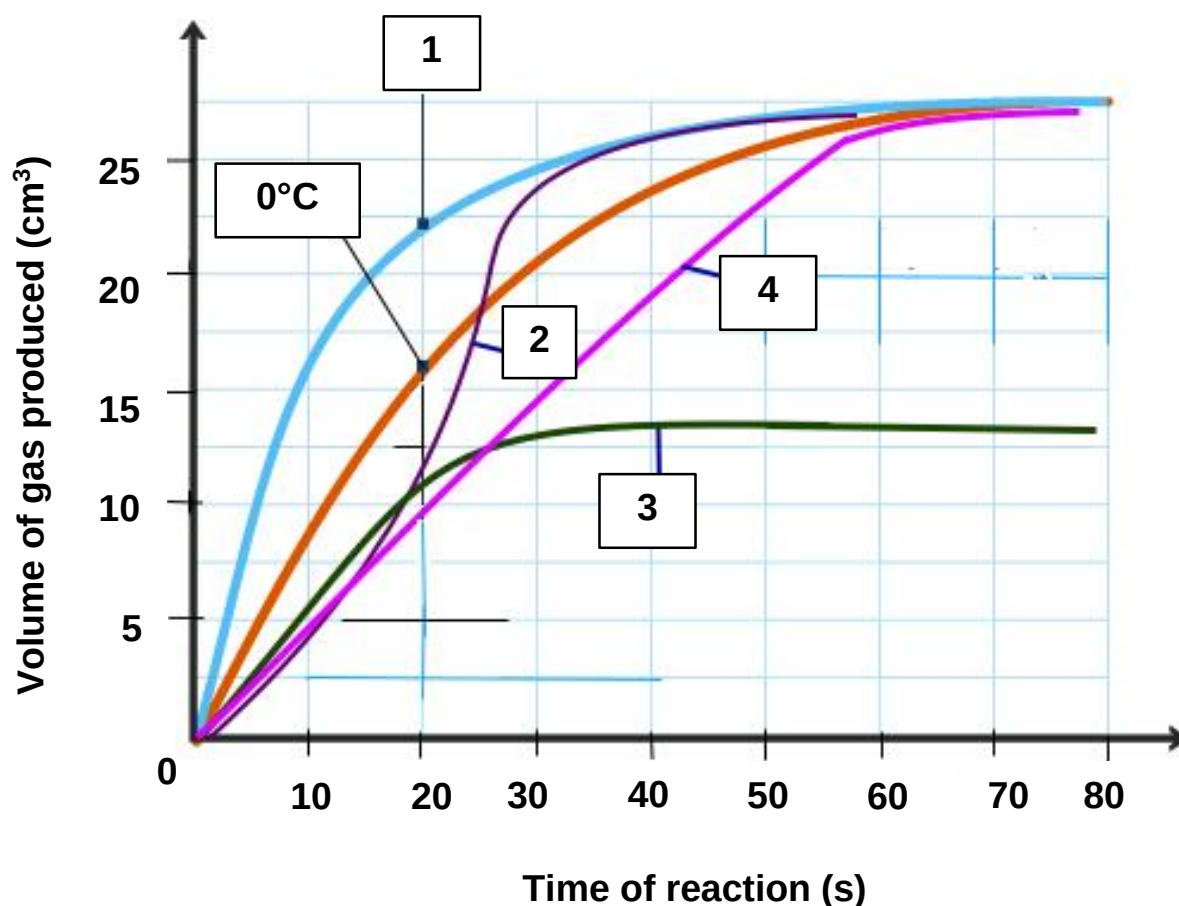
Question 1.1 - 1.4 refer to the information and graph below:

Blade investigated the rate of the reaction between zinc powder and nitric acid. The balanced chemical equation is shown below:



He initially carries out the reaction at 0°C, and at a pressure of 1 atm. He then repeats the experiment at 10°C at the same pressure. In all experiments he uses a 1,0 M solution of acid that is in excess.

The graph below illustrates the data that he collected during the course of the reaction.



1.1 Which one of the following apparatus would Blade have used to collect the data shown in the graph? (2)

- A. Scale
- B. Pressure sensor
- C. Syringe
- D. Spectrometer

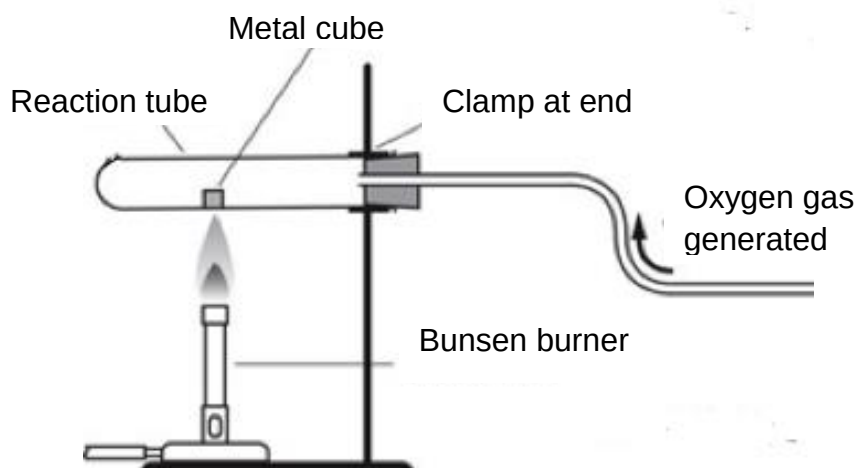
- 1.2 Which one of the curves labelled on the graph show the reaction that was carried out at 10°C? (2)
- A. 1
B. 2
C. 3
D. 4
- 1.3 What mass of zinc (in grams) did Blade use when the temperature was 0°C ? (2)
- A. 0,09 g
B. 0,17 g
C. 30,0 g
D. 87,1 g
- 1.4 Which one of the curves labelled show the reaction that took place when the same mass of zinc at 0°C was reacted with a nitric acid solution with a concentration of 0,5 M? (2)
- A. 1
B. 2
C. 3
D. 4
- 1.5 Which one of the following does NOT contain a dative covalent bond? (2)
- A. CO₂
B. SO₂
C. SO₃
D. NH₄⁺
- 1.6 What is the chemical formula for ammonium dichromate? (2)
- A. (NH₃)₂Cr₂O₇
B. (NH₄)₂Cr₂O₇
C. NH₄Cr₂O₇
D. NH₃Cr₂O₇
-

QUESTION 2: BONDING AND IMF'S

23 MARKS

Most metals react with oxygen upon heating. The ease with which the reaction takes place, as well as the vigor of the reaction can be used to determine which of the metals are most reactive.

Nathan performs a series of experiments to determine the order of reactivity of the group 1 elements. He uses lithium, sodium and potassium in his experiments. He connects the apparatus as shown below:



Nathan used the following method in each experiment:

1. Cut piece of Group 1 metal into a cube
2. Place metal into reaction tube
3. Connect reaction tube to conical flask where the oxygen is liberated.
4. Observe the ease at which the reaction takes place, and the vigor of the reaction.

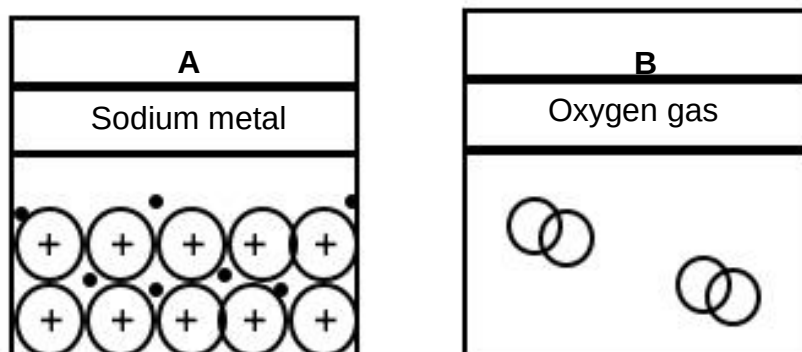
2.1 Describe one variable that Nathan should keep the same to ensure the experiment is a fair test. (2)

Nathan compiled the table below to summarize his results:

Metal	Observation
Lithium	Burns with a strongly red-tinged flame.
Sodium	Burns with a strong orange flame
Potassium	Melted to give a silver-like substance with no flame seen.

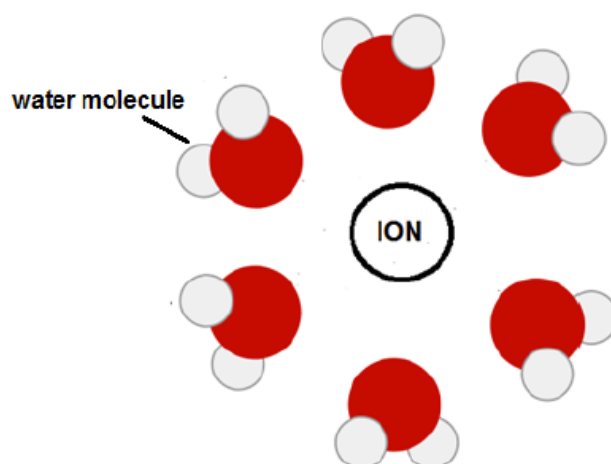
2.2 Explain why the data given in the table cannot be used to determine which metal was most reactive. (2)

- 2.3 Consider the reaction between lithium metal and oxygen to form lithium oxide.
- 2.3.1 Define the term *oxidation*. (2)
- 2.3.2 Write down the oxidation half reaction for the formation of lithium oxide. (2)
- 2.3.3 Give the net ionic reaction for this reaction. (3)
- 2.4 Consider the particle diagrams below (A and B) representing sodium metal and oxygen gas respectively:



- 2.4.1 Use diagram A to explain why sodium conducts electricity. (2)
- 2.4.2 Explain fully why sodium is a solid while oxygen is a gas at room temperature. Refer to intermolecular and intramolecular forces in your answer. (4)
- 2.5 Potassium burns in oxygen to form potassium oxide.
- 2.5.1 Write a dissociation reaction for the dissolution of potassium oxide (K₂O) in water. Include state symbols. (3)

Consider the diagram below showing how potassium oxide dissolves in water. The ion is surrounded by water molecules.



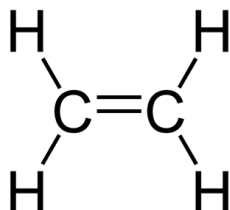
- 2.5.2 Is the ion shown in the diagram the potassium ion or the oxygen ion? Explain your answer fully. (3)

QUESTION 3: BONDING AND ENERGY

25 MARKS

Ethene (C₂H₄) is the most important organic chemical, by tonnage, that is manufactured.

Below is a Couper diagram of a molecule of ethene:



- 3.1 Draw a Lewis diagram to show the bonding in a molecule of ethene. (2)
- 3.2 Consider the C=C bond. On your Lewis diagram, label the pi and sigma bonds that occur between the carbon atoms. (2)
- 3.3 State the polarity of each of the following bonds and use a calculation to substantiate your answer.
- a) C – H bond (2)
- b) C = C bond (2)
- 3.4 Use your answer from 3.3, and other relevant information to explain fully whether a molecule of ethene would be polar or non-polar. (2)
- 3.5 High yields of bio-based ethene are obtained by dehydrating ethanol vapour using a catalyst:

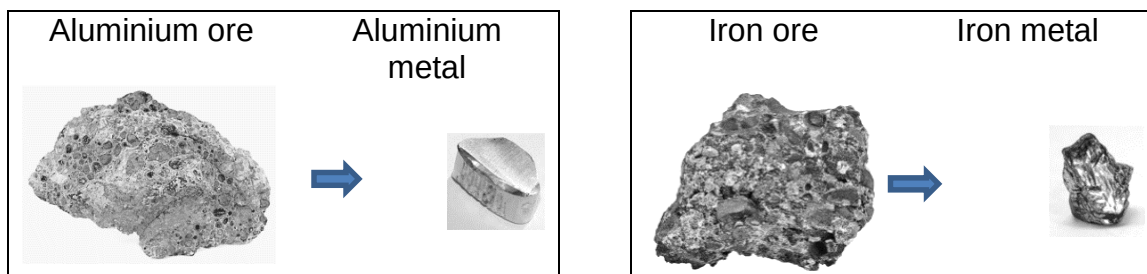


- 3.5.1 Define the term *activated complex*. (2)
- 3.5.2 Sketch a labelled energy profile diagram for the dehydration of ethanol (not to scale) *without* the use of a catalyst. Include the following labels in your diagram: (4)
- Activated complex
 - ΔH
- 3.5.3 Define the term *catalyst*. (2)
- 3.5.4 On your diagram, show the effect of using a catalyst using a dotted line (-----). (1)
- 3.5.5 Define the term *enthalpy change*. (2)
- 3.5.6 Is energy released or absorbed when the bonds within ethanol (C₂H₅OH) are broken? (1)
- 3.5.7 Is energy released or absorbed when the new bonds in ethene (C₂H₄) and water are formed? (1)
- 3.5.8 Use your answers from 3.5.6 and 3.5.7 to explain why the net enthalpy change for the dehydration of ethanol is positive. (2)

QUESTION 4: STOICHIOMETRY

27 MARKS

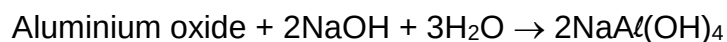
A student was asked to compare the industrial processes used to extract aluminium and iron from their ores.



Aluminium is extracted from purified aluminium oxide.

4.1 Give the chemical formula for aluminium oxide. (2)

The aluminium oxide reacts with hot concentrated sodium hydroxide and water to form a solution of sodium tetrahydroxoaluminate ($\text{NaAl}(\text{OH})_4$) according to the following balanced chemical equation:



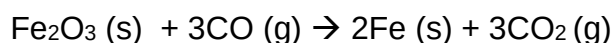
4.2 200 ml of a $2,0 \text{ mol} \cdot \text{dm}^{-3}$ solution of sodium hydroxide is reacted with excess aluminium oxide and 12 g of water:

4.2.1 Determine which reagent is limiting between the sodium hydroxide and the water. (4)

4.2.2 Explain the term *Yield* in the context of a chemical reaction. (2)

4.2.3 If 45 g of $\text{NaAl}(\text{OH})_4$ is produced in this reaction, determine the percentage yield of $\text{NaAl}(\text{OH})_4$. (5)

In the extraction of iron from its ore, carbon monoxide is reacted with iron oxide according to the following balanced chemical equation:



4.3 A mass of 1 kg of Fe must be produced. Assume the reaction is 100% efficient and the carbon monoxide is in excess.

4.3.1 Determine the amount of iron oxide that is required. (3)

4.3.2 Determine the number of iron atoms that is required. (4)

4.3.3 Determine the volume of carbon dioxide that is produced if the reaction is carried out at STP. (3)

4.3.4 What mass of iron oxide would be required if the reaction was only 70% efficient? 1 kg of Fe must still be obtained. (4)

QUESTION 5: STOICHIOMETRY

17 MARKS

Leroy preformed an experiment to determine the percentage of water of crystallization in hydrated sodium carbonate.

5.1 Explain what is meant by the term 'water of crystallization'.

(2)

The water of crystallization is normally expressed in the chemical formula of the substance. The formula for hydrated sodium carbonate is $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$.



5.2 Explain what is meant by the 'x' in the formula.

(2)

Leroy used a procedure known as a titration to determine the percentage of water of crystallization in sodium carbonate. The hydrated sodium carbonate is reacted with a solution of hydrochloric acid (HCl) of known concentration.

The first step in this procedure is to prepare a 2 M solution of HCl by diluting a concentrated HCl solution (10 M).

5.3 What is a solution of known concentration referred to as?

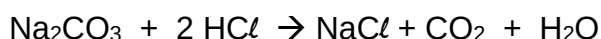
(1)

5.4 Determine the volume of water must be added to 200 ml of the 10 M HCl solution to prepare a solution of HCl with a concentration of 2 M.

(3)

Leroy reacted 23,5 ml of the 2 M hydrochloric with the hydrated sodium carbonate. All the dehydrated sodium carbonate reacts.

The balanced equation is shown below:



5.5 Determine the moles of dehydrated sodium carbonate that reacted.

(3)

Leroy correctly calculated that the mass of the dehydrated sodium carbonate that reacted is 2,491 g.

5.6 If the mass of the hydrated sodium carbonate was 5 g, determine the number of moles of water that is present in the water of crystallization.

(3)

5.7 Hence, determine the value of x in the formula $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$.

(3)

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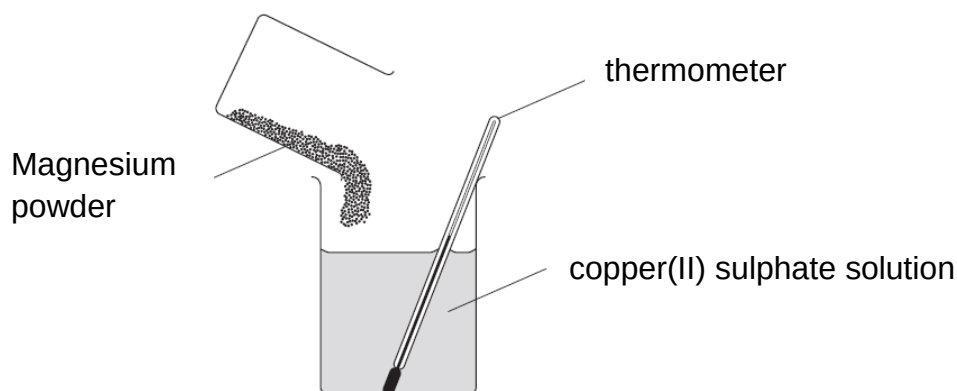
QUESTION 6: SCIENTIFIC SKILLS

26 MARKS

Katlego investigated the rate of the reaction between magnesium metal powder and copper(II) sulfate solution (CuSO_4).



He measured the temperature change when 0,5 g of magnesium was added to 50 ml of copper(II) sulfate solution. The change in temperature was used to determine the rate of the reaction.



He then repeated the experiment using different masses of magnesium powder. In each experiment he used 50 ml of copper(II) sulfate solution. He repeated the experiment for each mass of magnesium three times, and calculated an average. The table below summarizes his results:

Mass of magnesium powder (g)	Increase in temperature ($^{\circ}\text{C}$)			
	Trial 1	Trial 2	Trial 3	Average
0,5	1,5	2,0	2,5	2,0
1,0	3,5	4,0	5,0	4,2
1,5	6,0	8,0	5,0	6,3
2,0	6,0	6,5	6,0	6,2
2,5	5,0	7,0	6,0	6,0

- 6.1 Define the term *temperature*. (2)
- 6.2 Identify the indicating variable in Katlego's experiment. (2)
- 6.3 Katlego examined his results and concluded that the 2,0 g of magnesium gave the most accurate results. Comment on the validity of his statement. (2)

- 6.4 An incomplete graph of Katlego's results is shown on the ANSWER SHEET. Complete the graph by:
- 6.4.1 Providing the labels for the axis. (2)
 - 6.4.2 Providing a suitable scale for the y axis. (1)
 - 6.4.3 Plotting the missing values. (2)
 - 6.4.4 Drawing a best fit curve for the data. (2)
- 6.5 Use your graph to find the mass of magnesium required to produce a temperature rise of 3°C . Show how you obtained your answer by drawing two clear dotted lines on the grid on your answer sheet. (2)
- 6.6 Provide an explanation as to why the increase in temperature stays relatively constant for the last three mass readings (1,5 g, 2,0 g and 2,5 g). (2)
- 6.7 Explain fully why the temperature rise increased as the mass of the magnesium powder increased from 0 to 1,5 g. (4)
- 6.8 On the graph on the answer sheet, show the effect on the temperature rises if Katlego were to repeat the experiment using the same masses of zinc granules instead of zinc powder. **Label this new curve clearly 'Question 6.8'.** (2)
- 6.9 Explain your answer to 6.8. (3)
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END OF EXAM